IntelliVOD

Semantic Video Intelligence Platform

🧠 Project Overview

IntelliVOD is a **semantic video search engine** that allows developers to search for **exact technical explanations inside long-form videos (VODs)**.

Instead of watching an entire **2–3 hour conference talk or tutorial**, IntelliVOD identifies the **precise timestamp where a concept is explained**.

For example:

User Query:

```
How to implement 2FA in FastAPI?
```

Instead of returning the full video, IntelliVOD responds with:

```
Video: FastAPI Authentication Guide
Timestamp: 14:32
Explanation: "Here the instructor demonstrates implementing TOTP based two-factor authenticat
ion."
```

The platform essentially turns videos into **searchable knowledge bases**.

🎯 Core Objective

The goal of IntelliVOD is to enable developers to:

- Instantly find explanations inside long technical videos
- Skip irrelevant content
- Build a personal searchable knowledge library
- Combine documentation + video learning into one interface

This transforms traditional video learning platforms like **YouTube** into **semantic learning engines**.

🚀 Key Features

🔎 Timestamp-Level Semantic Search

Instead of searching **video titles or descriptions**, IntelliVOD searches **transcribed video content**.

The system can answer questions like:

- "What is event loop in Node.js?"
- "How does Redis caching work?"
- "How to implement OAuth in FastAPI?"

It returns:

- Video
- Timestamp
- Contextual transcript snippet

🌐 Internet-Wide Knowledge Expansion

If a query cannot be answered from the **local video library**, IntelliVOD automatically:

1. Searches the internet for relevant videos.
2. Downloads them using **yt-dlp**.
3. Transcribes them.
4. Indexes them into the vector database.
5. Returns timestamp-level results.

This creates a **self-growing knowledge engine**.

🧠 AI-Powered Video Understanding

Videos are processed using **speech-to-text transcription** powered by **OpenAI Whisper**.

This allows the system to understand:

- Spoken explanations
- Technical terminology
- Programming concepts

Once transcribed, the content becomes fully searchable.

📂 Intelligent Category Clustering

Videos are automatically categorized using **LLM-based tagging**.

Examples of categories include:

- DevOps
- System Design
- Backend Development
- Frontend Development
- Machine Learning

This makes the video library **self-organizing**.

📄 Multi-Modal Knowledge Search

IntelliVOD integrates with your **IntelliDoc system**, enabling users to search across:

- Video transcripts
- Technical documentation (PDFs)
- Knowledge notes

A single query can return results from **both videos and documents**.

Example query:

```
Explain distributed tracing in microservices
```

Results may include:

- Video timestamp explanation
- PDF documentation section
- Code snippet reference

🧠 System Pipeline

The IntelliVOD system follows a **multi-stage AI pipeline**.

1️⃣ Video Ingestion

The system downloads videos using:

- **yt-dlp**

Sources include:

- YouTube tutorials
- Twitch developer streams
- Technical conference recordings

The video metadata is stored in the database.

2️⃣ Speech Transcription

Videos are transcribed using:

- **OpenAI Whisper**

Output:

```
Timestamped transcript
```

Example:

```
00:02:12 - Today we will implement JWT authentication in FastAPI.
00:04:33 - Now let's add the login endpoint.
```

3️⃣ Transcript Chunking

Large transcripts are split into **semantic chunks**.

Example chunk:

```
Chunk Size: 300–500 tokens
```

Each chunk retains its:

- Video ID
- Timestamp
- Context window

4️⃣ Embedding Generation

Each transcript chunk is converted into a **vector embedding** using:

- **OpenAI text-embedding-3-small**

These embeddings represent the **semantic meaning of the text**.

5️⃣ Vector Storage

Embeddings are stored in a vector database such as:

- **ChromaDB**
- **Pinecone**

This enables fast similarity search.

6️⃣ Semantic Retrieval

When a user submits a query:

1. The query is converted into an embedding.
2. Vector similarity search is performed.
3. Relevant transcript chunks are retrieved.

Similarity is measured using **Cosine Similarity**.

7️⃣ Timestamp Resolution

Once the best transcript match is found:

The system extracts:

- Video ID
- Timestamp
- Transcript snippet

The frontend then jumps **directly to that moment in the video**.

🏗️ System Architecture

The platform uses a **modern AI-driven architecture**.

Component	Technology	Responsibility
Frontend	Next.js + Tailwind	UI and video interface
Backend API	FastAPI	Query handling and ML pipeline
Vector Database	ChromaDB / Pinecone	Semantic search
Metadata DB	PostgreSQL	Video metadata storage
Scraper Worker	Python Workers	Video ingestion and processing
Transcription Engine	Whisper	Speech-to-text conversion
Orchestration	Redis Queue	Background job management

⚙️ Execution Workflow

1️⃣ User Query

A user searches:

```
How does OAuth work in FastAPI?
```

2️⃣ Embedding Creation

The query is converted into a vector representation.

3️⃣ Vector Search

The system searches the vector database for **similar transcript chunks**.

4️⃣ Timestamp Match

The most relevant transcript snippet is identified along with its **video timestamp**.

5️⃣ Result Delivery

The frontend displays:

- Video player

- Timestamp jump
- Transcript snippet

Example:

```
Video: FastAPI Security Tutorial
Timestamp: 08:41
Explanation: OAuth authentication flow
```

📊 Data Storage Architecture

The system uses a **hybrid database model**.

PostgreSQL (Structured Data)

Stores:

- Video metadata
- Channel information
- Category tags
- Transcript references

Vector Database

Stores:

- Transcript embeddings
- Semantic index

Used for **similarity search queries**.

🛠️ Recommended Tech Stack

To maximize development efficiency and production readiness.

Frontend

- Next.js
- Tailwind CSS
- React Player
- TypeScript

Backend

- **FastAPI**

Handles:

- Semantic search API
- Query processing
- Worker orchestration

Databases

Structured Data

- **PostgreSQL**

Vector Storage

- **ChromaDB**
- **Pinecone**

DevOps

Containerization:

- **Docker**

Deployment:

- **Amazon EC2**

CI/CD:

- **GitHub Actions**

🔮 Future Enhancements

🎥 Real-Time Video Indexing

Automatically index newly uploaded videos from developer channels.

🧠 AI Learning Assistant

Allow users to ask follow-up questions about a video segment.

Example:

```
Explain this part more simply
```

📚 Personal Knowledge Vault

Users can upload:

- Courses
- Conference recordings
- Lecture videos

to create a **personal semantic learning library**.

👫 Collaborative Knowledge Graph

Developers can share curated knowledge graphs of technical videos.

📌 Why This Project Matters

Technical knowledge today is heavily stored in **video content**, but traditional video platforms are not designed for **precise knowledge retrieval**.

IntelliVOD converts videos into:

- searchable data
- semantic knowledge
- timestamp-level insights

This makes long-form video learning **as searchable as documentation**.